

control. More important, it suggests that they would have understood the importance of enduring some immediate (and excruciating) pain to reduce the chance of death. Such a calculation would have been impossible without causal reasoning, an attribute whose presence among Neanderthals has long been debated. This finding helps to bring that debate to a close.

“It’s pretty exciting to see such ancient evidence of a medical procedure as complex as this,” says Daniel Lieberman, an evolutionary biologist at Harvard University. “Sometimes I joke when I’m teaching that humans are the only species that would ever willingly go to a dentist. Now,” he adds, “I’ll have to revise my quip.” ■

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Science & technology | Cloudy, with a chance of revolt

AI models are being used to predict conflict

Good data are hard to come by

May 14th 2026



AS AN UNEASY truce holds between America and Iran, experts are struggling to predict what new phase the conflict may enter next. Might an artificial-intelligence model know any better? To find out, The Economist asked RAND, a think-tank, to see if its new AI forecasting system thought a pop-

ular uprising was in the offing in Iran. Integrated Strategic Forecasting (ISF), as the system is known, put the chance of regime collapse or replacement by the end of 2026 at 20%—higher than many experts would hazard.

There are caveats. The forecast was produced without classified intelligence. Its inputs and outputs were also not vetted by humans, as is customary for forecasts commissioned by government agencies, notes Anthony Vassalo, RAND's head of prediction technology and a former senior official in America's Office of the Director of National Intelligence. Even so, Mr Vassalo is bullish. He describes ISF, which was completed in February, as "the better crystal ball" policymakers have long sought.

And ISF is not the only game in town. After two decades in which attempts to build conflict-predicting computer models have yielded disappointing results, recent advances in machine learning and large language models have prompted many data scientists to take another crack.

The idea is simple. Models trained on past conflicts are fed indicators that may signal future strife, in the hope that predictive patterns invisible to humans will emerge. Inputs include data on

crime, public health, labour strikes, weather, the economy and political developments such as democratic backsliding. Social media are widely mined to gauge discontent.

Forecasters are also turning to images from satellites, drones and surveillance cameras. The aim is to spot clues in street life, traffic patterns, and the way protesters converge, hold ground and disperse. Such image analysis is being incorporated into a "prediction hub" at the University of the German Armed Forces in Munich. The system will serve Germany's defence ministry, says Daniel Racek, the project's chief early-warning data scientist.

Useful as all this may be, the best predictor of conflict is past conflict, says Katayoun Kishi, chief data scientist at ACLED, a non-profit in Wisconsin. ACLED pays some 150 researchers worldwide to track riots, government crackdowns, gang warfare, military attacks and other violence that raise the odds of future conflict. These data—deemed complete enough to be used by ForecastBench, a non-profit, to assess general-purpose forecasting models—are fed into ACLED's model, CAST.

CAST then uses this information, augmented with indicators such as infant mortality and occur-

rences of peace talks, to predict bouts of organised political violence up to six months out. Where good data exist, says Dr Kishi, the model works well. CAST correctly predicted, for instance, that in July 2023 the Brazilian state of Ceará, which had seen fighting among criminal factions, would have two battles, four attacks on civilians and no bombings. CAST, which has since been enhanced with reporting on other countries, is used by UN agencies and the Dutch foreign ministry.

Models are also getting better at assessing “risk amplifiers”. Heatwaves, for instance, have long been linked to an increased likelihood of riots. Proximity to extractable resources also matters, says Havard Hegre, who has trained NATO officials on an AI forecasting model, Violence & Impacts Early-Warning System (VIEWS), that he leads at the Peace Research Institute Oslo. Where there is oil or diamonds, successful rebels can expect to cash in, increasing the odds of conflict. VIEWS’s insights have led to its adoption by the UN Development Programme (UNDP) and the EU’s diplomatic service.

One newly upgraded set of models, run by the International Organisation for Migration (IOM), a

UN body, aims to predict how many people will be displaced by conflict and natural disasters. The system’s first forecast, published on April 1st, predicted with ambitious precision that drought, floods and fighting would drive 304,362 people in Somalia from their homes over the next three months. As of April 20th Eva Nyaga, a data scientist at the IOM’s office in Nairobi, says the forecast may well be only a few percent off.

Scepticism remains, even among some users. Corrado Scognamillo of the UNDP’s crisis-risk unit in New York says his team considers just three publicly available models, including CAST and VIEWS, useful, but only if strife is already under way. When it comes to the trickier matter of predicting the onset of a new conflict, he says, all models appear unreliable. Available data, he argues, often fail to capture a crisis’s true triggers. Growing inequality, for instance, might prove less combustible than a change in how it is perceived.

To complicate matters, disinformation campaigns, now rampant, are muddying social media’s predictive value, says Jack Rooney, boss of Aldebaran Threat Consultants, an intelligence firm in Dubai. Many hope to reduce errors by casting a wider net. Aldebaran taps a network of human sources

to compile databases that some defence-ministry clients plan to feed into their own forecasting models. The Soufan Centre, a New York non-profit that also collects data for conflict forecasting, works to obtain undercover access to group chats run by outfits that advocate political violence. This allows them to pick up sentiments rarely expressed in public, says Clara Broekaert, a researcher at the centre.

History is, of course, littered with the wreckage of outfits with misplaced confidence in their predictions. Mr Scognamillo highlights another pitfall. Predictions of unrest, merited or not, will tempt some regimes to crack down pre-emptively, he says. They may also draw unwelcome attention to the newly vulnerable. Tellingly, the IOM published its Somalia forecast at reduced spatial resolution.

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Science & technology | Well Informed

Do houseplants improve air quality?

Air purifiers are more effective and easier to maintain

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Clean air is important for health, yet many homes are rife with pollutants. Household products such as cleaning solvents, waxes, paints and varnishes often contain volatile organic compounds like benzene and toluene, which can cause skin irritation, eye damage, neurological disorders and can-